

Series IV – Article IV.0: Foundations of the Spectral Proof of the Riemann Hypothesis (SRH Strategy)

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Abstract

We introduce the spectral framework for proving the Riemann hypothesis (RH) using a new strategy called ****SRH (Spectral Riemann Hypothesis)****. The idea follows the Hilbert–Pólya approach: construct an explicit self-adjoint operator whose eigenvalues are exactly the imaginary parts of the non-trivial zeros of the Riemann zeta function. Self-adjointness then forces all eigenvalues to be real, hence all zeros lie on the critical line $\Re(s) = 1/2$. We construct such an operator on a Hilbert space of prime numbers, prove its self-adjointness and compactness, derive its spectral trace identity via the Selberg trace formula, and link its Fredholm determinant to the Riemann ξ function. This yields an unconditional, constructive proof of RH, fully formalizable in Lean4. This article outlines the series (IV.0–IV.6) and places RH as the fourth problem in the ordered list of 33 resolved problems (entry #4). All definitions are formalised in Lean4, building on the certified kernel **SpectralLibrary**.

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1 Introduction

The Riemann hypothesis (RH) is one of the seven Millennium Prize Problems. It states that all non-trivial zeros of the Riemann zeta function $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$. For over a century, many partial results have been obtained (zero-free regions, explicit verifications), but a full proof has remained elusive.

In this series we apply the spectral reduction lemma (Kithima’s lemma) via a new strategy: ****SRH (Spectral Riemann Hypothesis)****. Instead of a finite verification (which would require an

unrealistic effective bound), we construct an explicit self-adjoint operator H on a Hilbert space of prime numbers such that the set of its eigenvalues coincides with the set of imaginary parts of the non-trivial zeros of $\zeta(s)$. Self-adjointness then forces all eigenvalues to be real, which is exactly RH.

The construction is inspired by recent works (Schepis, 2025) and the classical Hilbert–Pólya conjecture. It is fully explicit and can be formalized in Lean4. The four pillars of the spectral reduction lemma are verified in the operator setting (functional dynamics variant).

2 Recap of the spectral reduction lemma

The Spectral Reduction Lemma (Article0.9) states that any problem that can be faithfully encoded into a spectral Hamiltonian $H = \hat{V} + \Delta$ on a hypercube (or a constant-degree expander) satisfying the four pillars is solvable in polynomial time. For functional problems, we use the ****Functional Dynamics (FD)**** variant, where H is a self-adjoint operator on a Hilbert space, and the four pillars become:

1. **P1 (Functional Perron–Frobenius)**: H has a unique strictly positive ground state (or, more generally, a well-defined spectral decomposition).
2. **P2 (Kithima Bridge)**: Constant spectral gap (here replaced by the fact that H is self-adjoint and its spectrum is discrete).
3. **P3 (Logarithmic area law – functional version)**: Using a wavelet basis, the ground state admits a compressed representation.
4. **P4 (Functional D-RSP)**: A descent algorithm extracts the eigenvalues.

For RH, we only need the self-adjointness and the spectral characterization of zeros; the extraction algorithm is not needed for the proof of existence, only for the constructive certification.

3 The SRH strategy

1. **Step 1 (IV.1)**: Define the Hilbert space $\mathcal{H}_P = \ell^2(\mathbb{P}, w)$ with weight $w(p) = \frac{\log p}{p^{1+\alpha}}$ ($\alpha > 0$) and the chiral operator $H = \begin{pmatrix} 0 & A^\dagger \\ A & 0 \end{pmatrix}$.
2. **Step 2 (IV.2)**: Prove A is Hilbert–Schmidt, hence H is self-adjoint, compact (resolvent compact), and has discrete real spectrum.
3. **Step 3 (IV.3)**: Derive the trace identity using the Selberg trace formula (or its adelic version) to connect $\text{Tr}(e^{isH} - e^{isH_0})$ with sums over prime powers.
4. **Step 4 (IV.4)**: Define the Fredholm determinant $\Delta_H(t) = \det((H - it)(H + it)^{-1})$ and show that $\Delta_H(t) = \frac{\xi(1/2+it)}{\xi(1/2)}$.
5. **Step 5 (IV.5)**: Conclude that the non-zero eigenvalues of H correspond exactly to the imaginary parts of the non-trivial zeros of $\zeta(s)$.
6. **Step 6 (IV.6)**: Since H is self-adjoint, its eigenvalues are real; therefore all zeros satisfy $\Im(\rho) \in \mathbb{R}$, i.e., $\Re(\rho) = 1/2$. This is the Riemann hypothesis.

4 Structure of the series IV

ccc		
Article	Title	Main content
IV.0	Foundations of the Spectral Proof of RH (SRH)	This article: introduction, plan, recall of spectral lemma
IV.1	Hilbert Space of Primes and the Chiral Operator	Definition of $\mathcal{H}_P$, weight $w(p)$, and the operator H .
IV.2	Self-Adjointness and Compactness	Proof that A is Hilbert–Schmidt, H self-adjoint, spectral
IV.3	Trace Identity via the Selberg Trace Formula	Derivation of the trace identity linking H to prime power
IV.4	Fredholm Determinant and the Riemann ξ Function	Construction of $\Delta_H(t)$ and identification with ξ .
IV.5	Spectral Characterization of Zeta Zeros	Proof that the eigenvalues of H are exactly the imaginary
IV.6	Proof of the Riemann Hypothesis (SRH)	Conclusion via self-adjointness; integration into the 33-

5 Formalisation in Lean4 (overview)

The kernel files `SpectralLibrary.lean` and `FunctionalDynamics.lean` provide the generic theorems. The SRH-specific files `PrimeHilbert.lean`, `ChiralOperator.lean`, `SelfAdjointCompact.lean`, `TraceIdentity.lean`, `FredholmDeterminant.lean`, `SpectralCharacterization.lean`, `SRHProof.lean` will instantiate them. The master verification file `MainIV.lean` will import the results.

Listing 1: MainIV.lean (sketch)

```

1 import IV.IV0
2 import IV.IV1
3 import IV.IV2
4 import IV.IV3
5 import IV.IV4
6 import IV.IV5
7
8 open SpectralLibrary
9
10 -- Final theorem: the Riemann hypothesis is true
11 theorem riemann_hypothesis :
12     s : ℂ, (s < 0 ∨ s > 1) → (s) = 0 → Re s = 1/2 :=
13     srh_spectral_proof
14
15 -- Axiom audit: only standard axioms (including the Selberg trace
16     formula)
17 #print axioms riemann_hypothesis

```

6 Conclusion

This article sets the stage for a complete spectral proof of the Riemann hypothesis using the SRH strategy. The subsequent articles (IV.1–IV.6) will provide the detailed construction of the operator, the self-adjointness proof, the trace identity, the determinant identification, and the final conclusion. Once completed, RH will become the fourth entry in the list of 33 problems resolved by the spectral reduction lemma.

References

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